The shipped crack detector

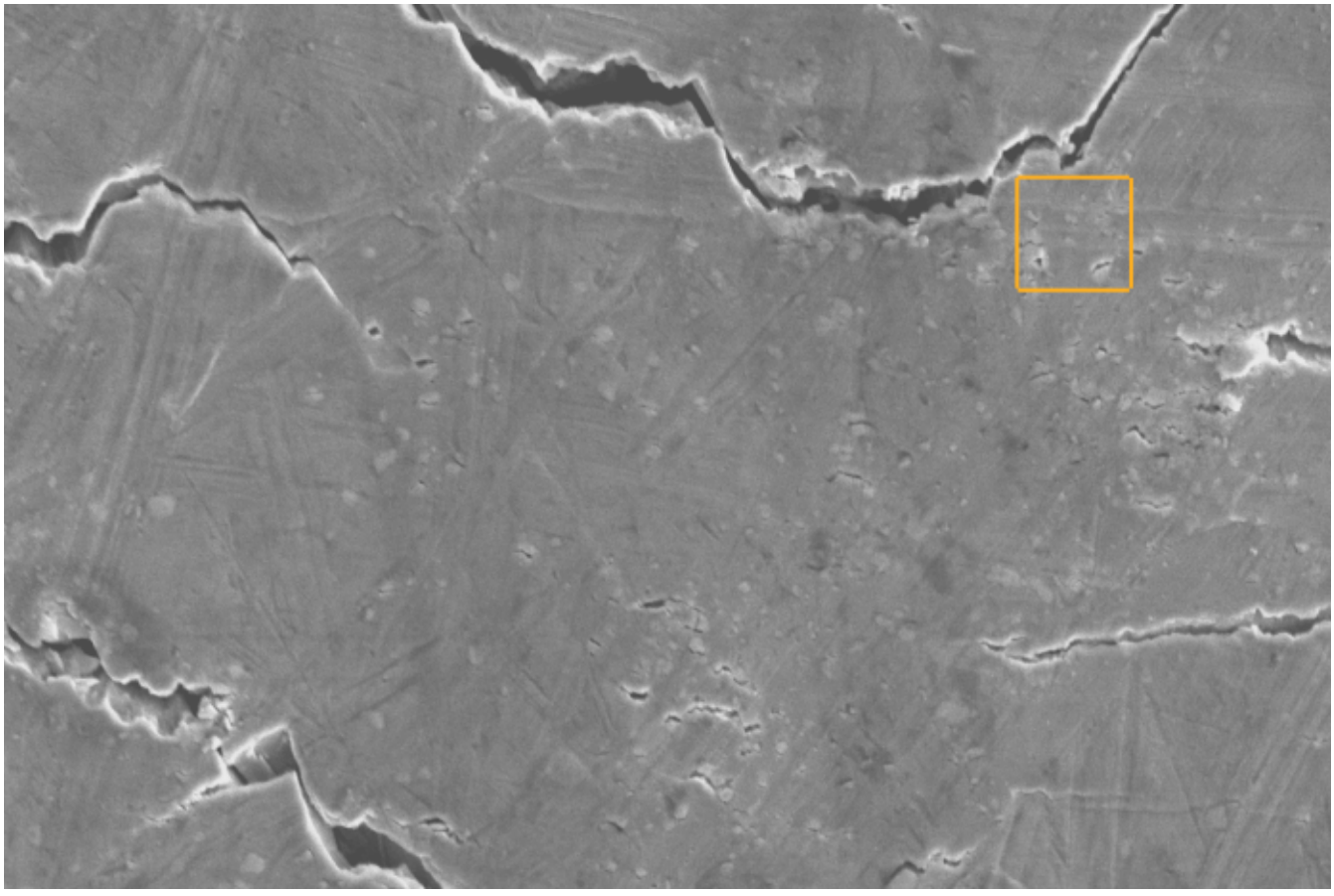
one real image per stage

Frame HIP_24hr_SE_Side_006. Window 262,144 px of the frame's 25,165,824. Every panel is a real array from the real pipeline: no mock-ups, no schematic boxes standing in for an image.

Why this frame. lower of the two central tiles by IoU over the 26 hand-labelled tiles (0.4039 and 0.4339 straddle the median 0.4189 at equal distance; the tie is broken downward so the page cannot flatter the detector)

Configuration: mad_k 5.0, absolute_dark_thresh 10, ladder (5,10,20,40,80,160,320), decimate above sigma 64, clean_mask min_area 13, region floor 40 px, USE_REGION_CLASSIFIER False.

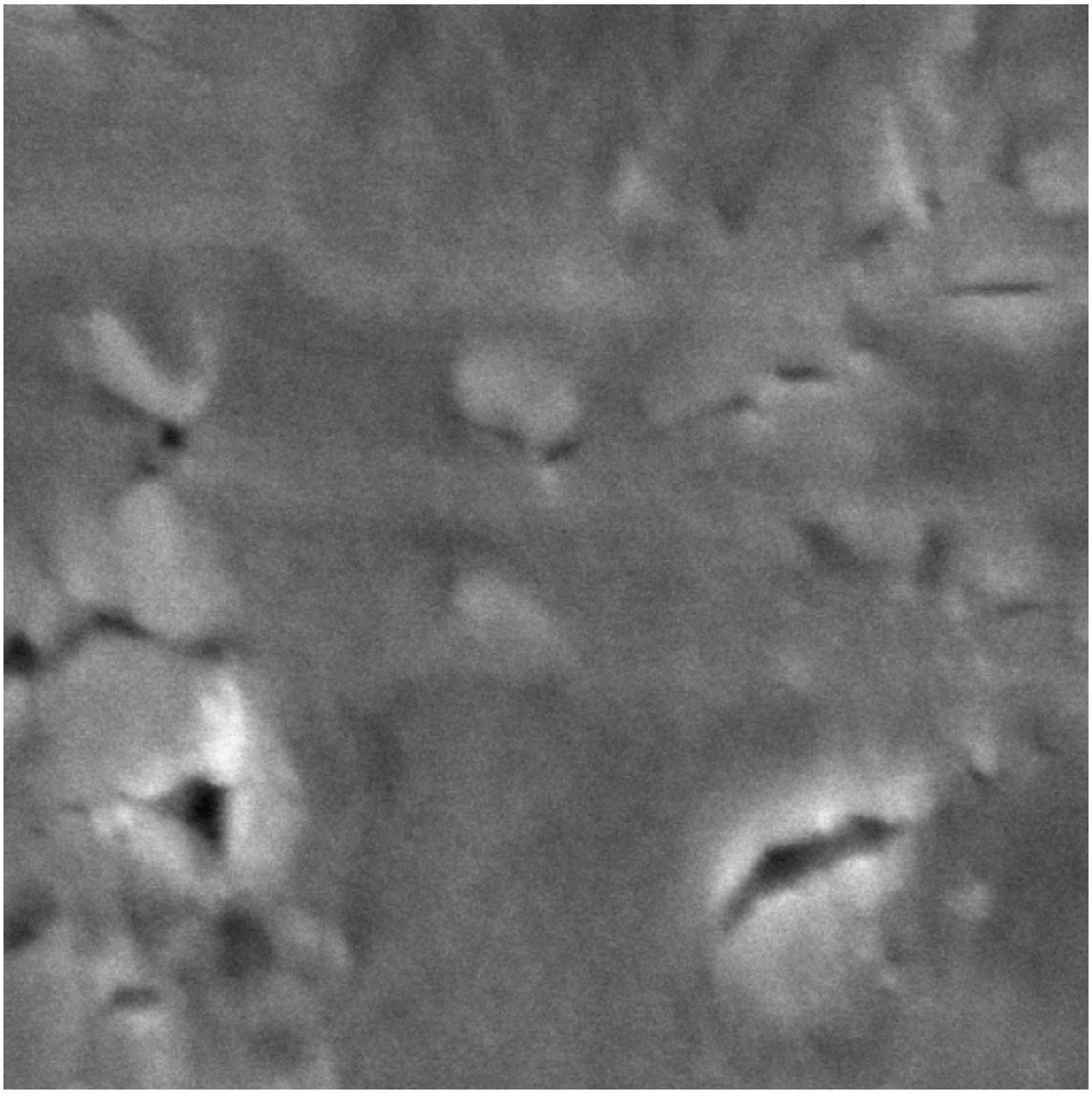
The whole frame; the window every other panel shows is marked



1. The micrograph

What it does. Reads the 16-bit TIFF, stretches it to 8 bit on its 0.5/99.5 percentiles, and crops the databar off: 6144×4096 px survive.

Cost. A percentile stretch, not min/max: a hot pixel or the crack itself at 0 would otherwise own the range and flatten the background texture Otsu needs downstream.

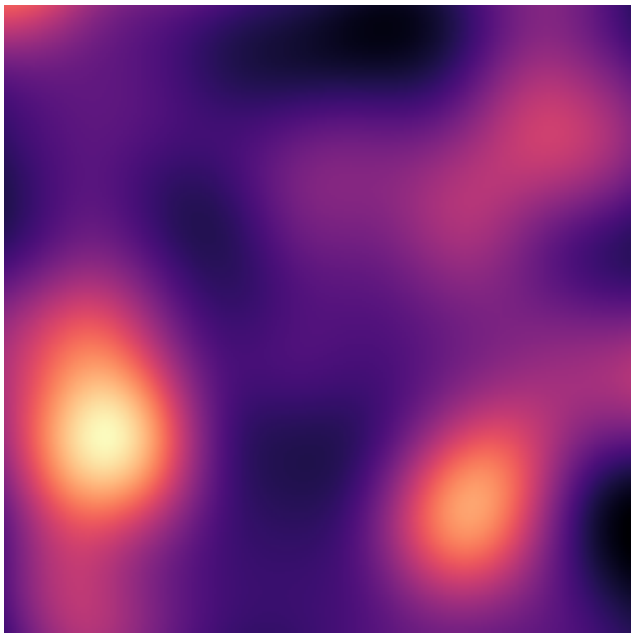


8-bit, after the 0.5/99.5 percentile stretch and the field-of-view crop.

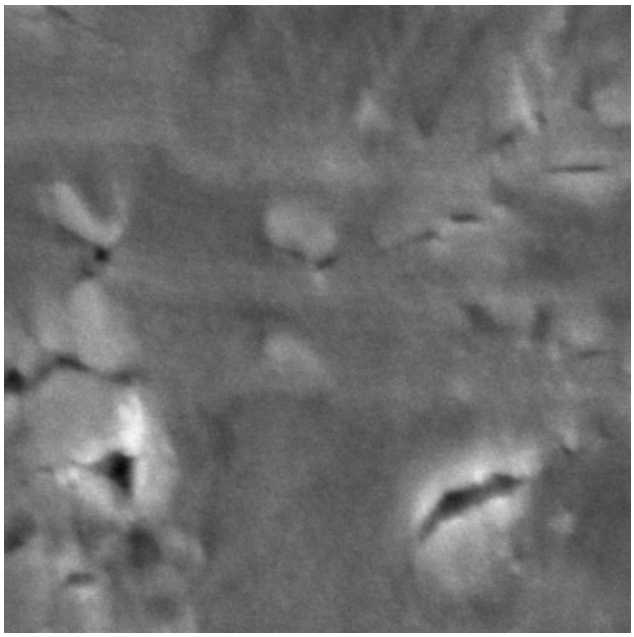
2. Why the next step exists: the illumination

What it does. Nothing yet — this is the quantity flatten_background is about to subtract.

Why it matters. A single global threshold on the raw image would track this drift instead of the crack.



The Gaussian background estimate at sigma=40 over this crop, false-coloured. It spans 92.8 to 139.3 grey levels across 512 px — drift of that size sits on top of every crack.

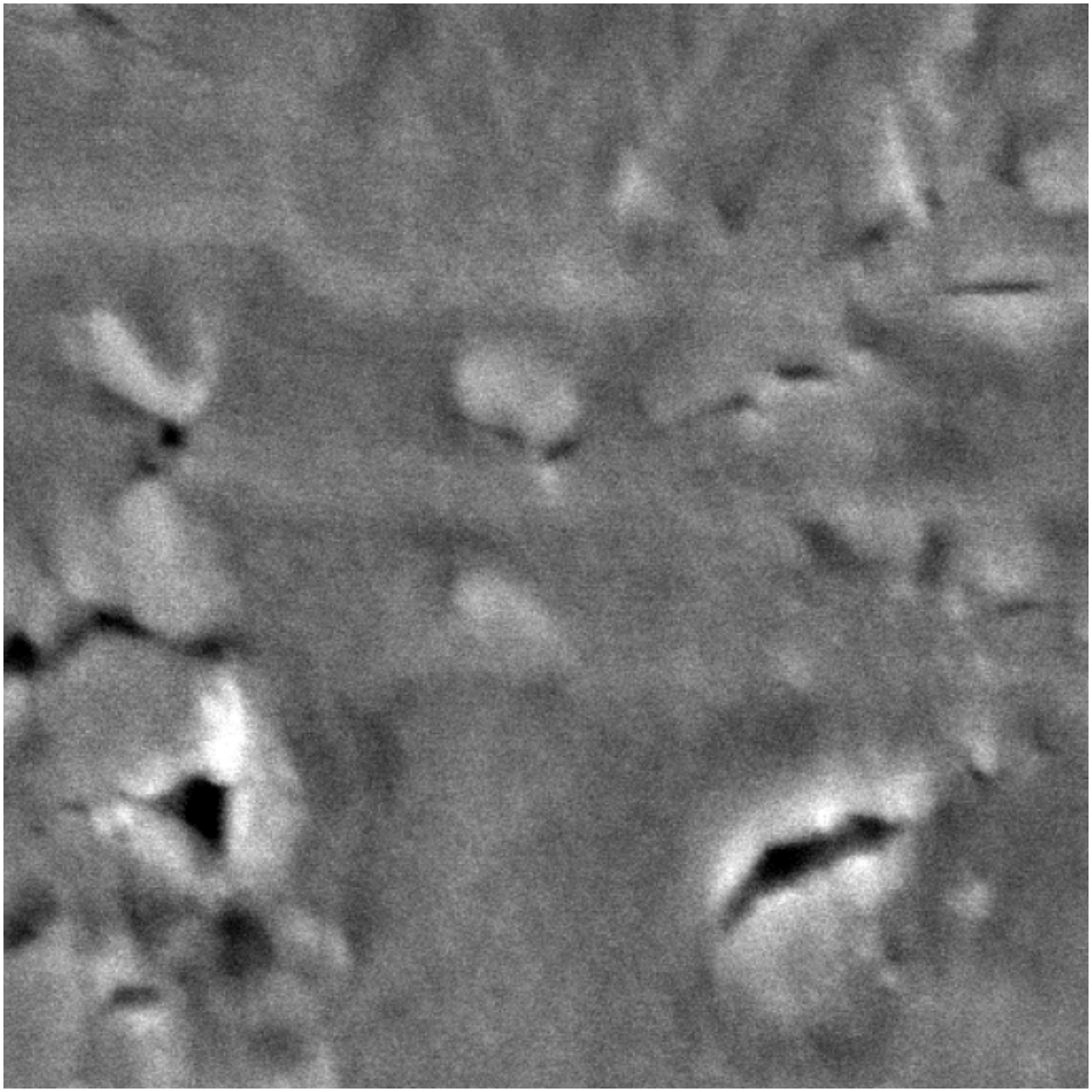


The same crop, as captured, for comparison.

3. flatten_background

What it does. Subtracts the blurred copy, so what is left is local contrast.

What it costs. It is a high-pass filter, so it also erases anything wider than its own radius. The detector's own docstring measures this: pixels that are genuinely pitch black (raw 0–4) come out near 122 of 255 after flattening, indistinguishable from mid-grey matrix. That single sentence is why steps 5 and 6 exist.



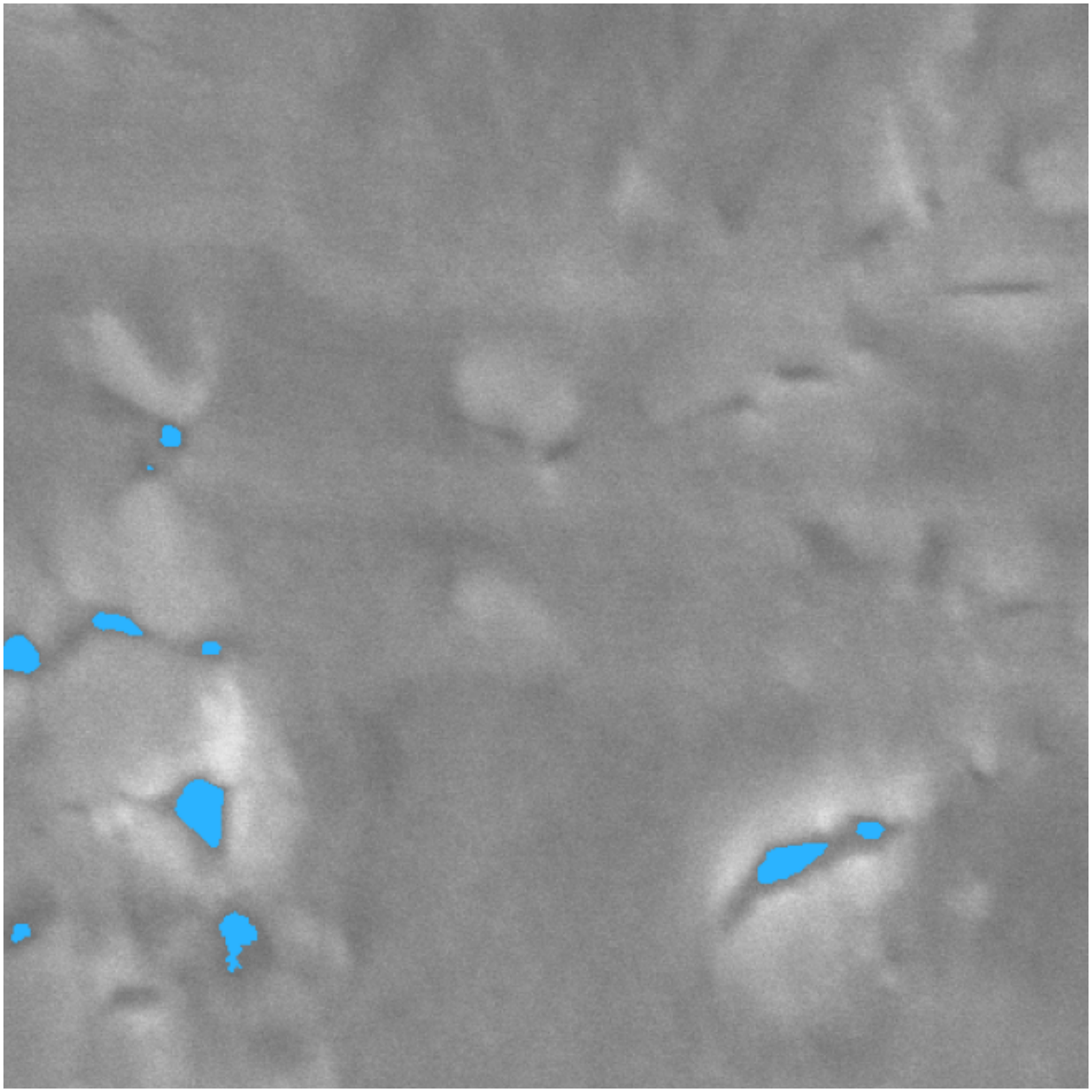
The flattened image: raw minus the background above, rescaled to 8 bit on its own 0.5/99.5 percentiles.

4. The darkness cut, at one radius

What it does. Keeps a pixel if the denoised flat image is below min(Otsu, median − 5×MAD), OR if the raw 8-bit image is below 10. Taking the lower of Otsu and the robust bar is what stops Otsu flagging half the frame on a unimodal histogram.

What it buys, on this crop. precision 0.9900, recall 0.2587, IoU 0.2580 against the hand paint. Almost everything it marks is crack; it finds about a quarter of it.

The absolute-darkness term alone carries 15 px here and 233,102 px frame-wide. At corpus scale it supplies roughly half of all the recall the detector has, and its threshold of 10 has no provenance; raising it is additive but three instruments disagree about whether to, so it stays put.

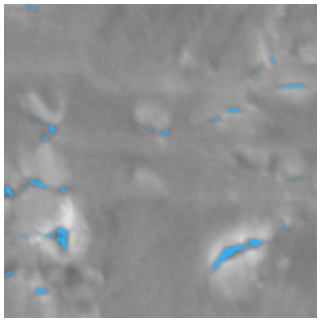


carries 1,808 px in this crop (0.69% of it) ·
483,991 px in the whole frame

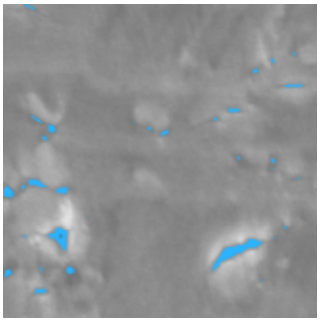
5. The same cut at each of the seven radii

What it does. Re-flattens the image at sigma = 5, 10, 20, 40, 80, 160, 320 and takes the identical cut at each. A crack is only reliably dark at the scales near its own width, so one radius sees one band.

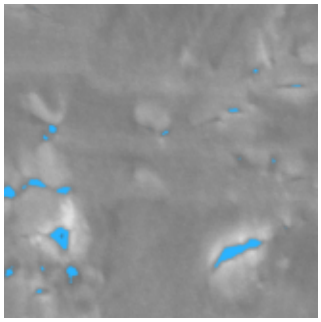
“New” counts pixels no earlier rung and not the absolute-darkness term had already claimed, taken in ascending sigma, so the seven new-counts sum to the union and no pixel is counted twice. Each rung is produced by calling the installed detector with that one rung in its ladder; the OR of the seven plus the absolute term is asserted equal, bit for bit, to the mask the pipeline itself returned.



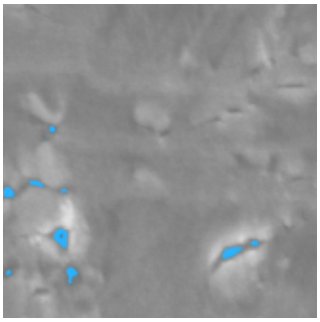
sigma = 5 · 2,155 px here · 2,155 of them new · 89,464 px frame-wide · 89,464 new frame-wide



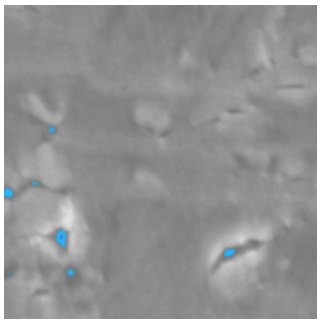
sigma = 10 · 3,456 px here · 1,382 of them new · 182,900 px frame-wide · 97,072 new frame-wide



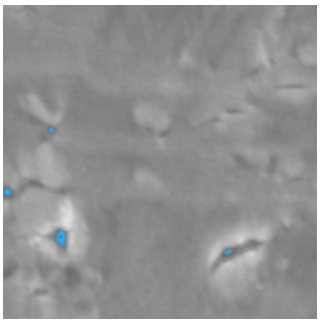
sigma = 20 · 3,012 px here · 292 of them new · 236,262 px frame-wide · 83,449 new frame-wide



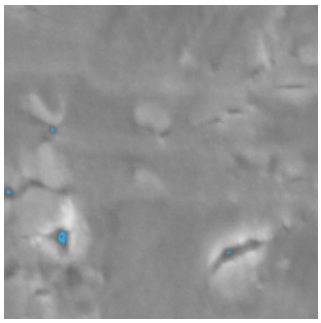
sigma = 40 · 1,793 px here · 45 of them new · 250,889 px frame-wide · 69,784 new frame-wide



sigma = 80 · 823 px here · 0 of them new · 266,073 px frame-wide · 71,974 new frame-wide



sigma = 160 · 400 px here · 0 of them new · 237,561 px frame-wide · 22,075 new frame-wide



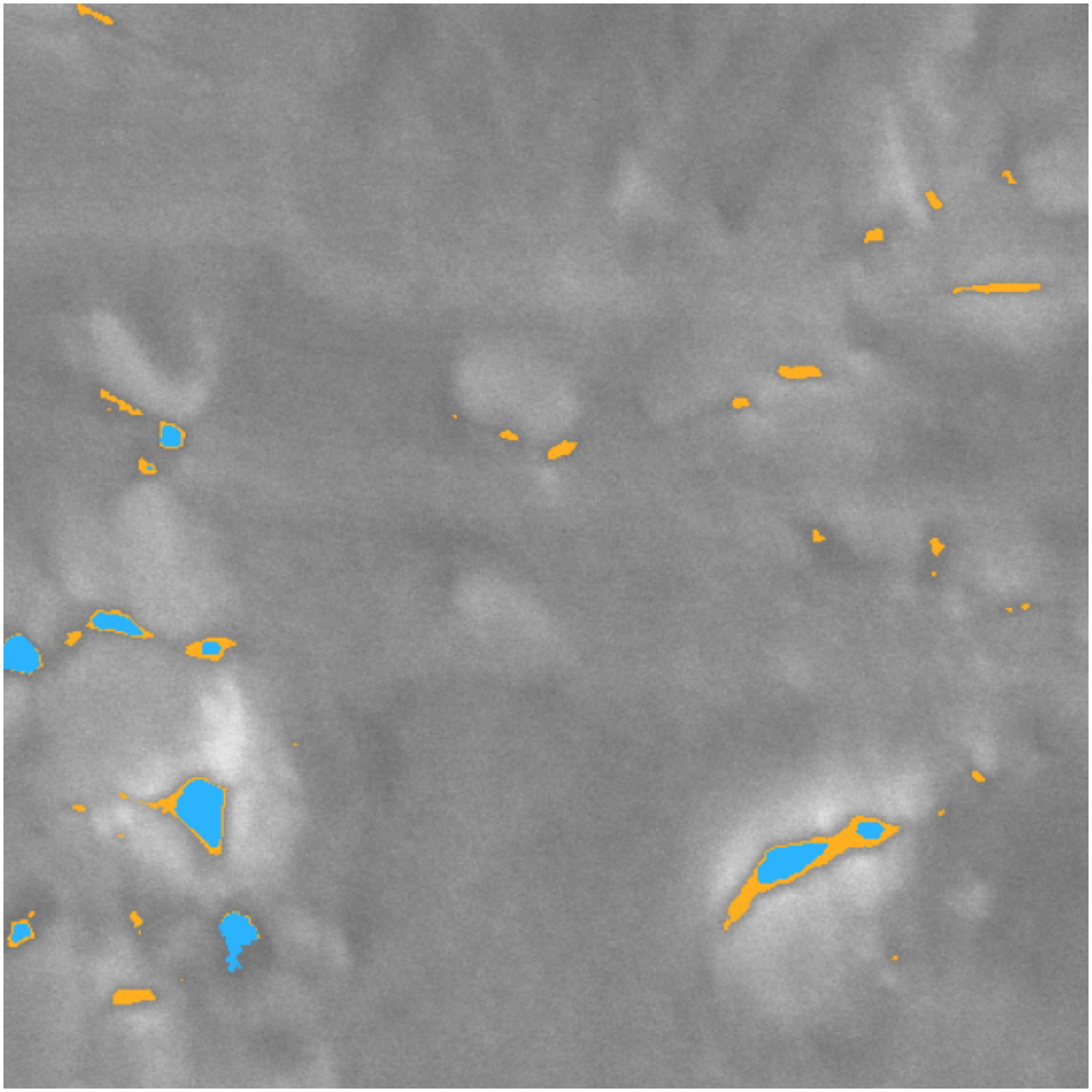
sigma = 320 · 230 px here · 0 of them new · 192,566 px frame-wide · 1,015 new frame-wide

6. Their union — the change v1.49 made

What it does. ORs the seven cuts. It is a strict superset of the single-radius mask by construction, so it cannot drop a pixel the older build kept — checked, not assumed: 0 px lost across the 26 labelled tiles and across 6,669,145 hand-painted pixels on 12 full frames.

What it buys, on this crop. +2,081 px here and +183,944 px frame-wide. precision 0.9900 → 0.8138, recall 0.2587 → 0.4574, IoU 0.2580 → 0.4141. Over the 26 labelled tiles the same change reads IoU 0.2979 → 0.3344 pooled, and it costs no new tuned constant: the ladder is every scale, not a chosen subset.

What it costs. Seven flattening passes instead of one: 12.7 s of this frame's 130.2 s. Above sigma=64 the background is estimated on a decimated copy, which is what keeps that from being 67 s.

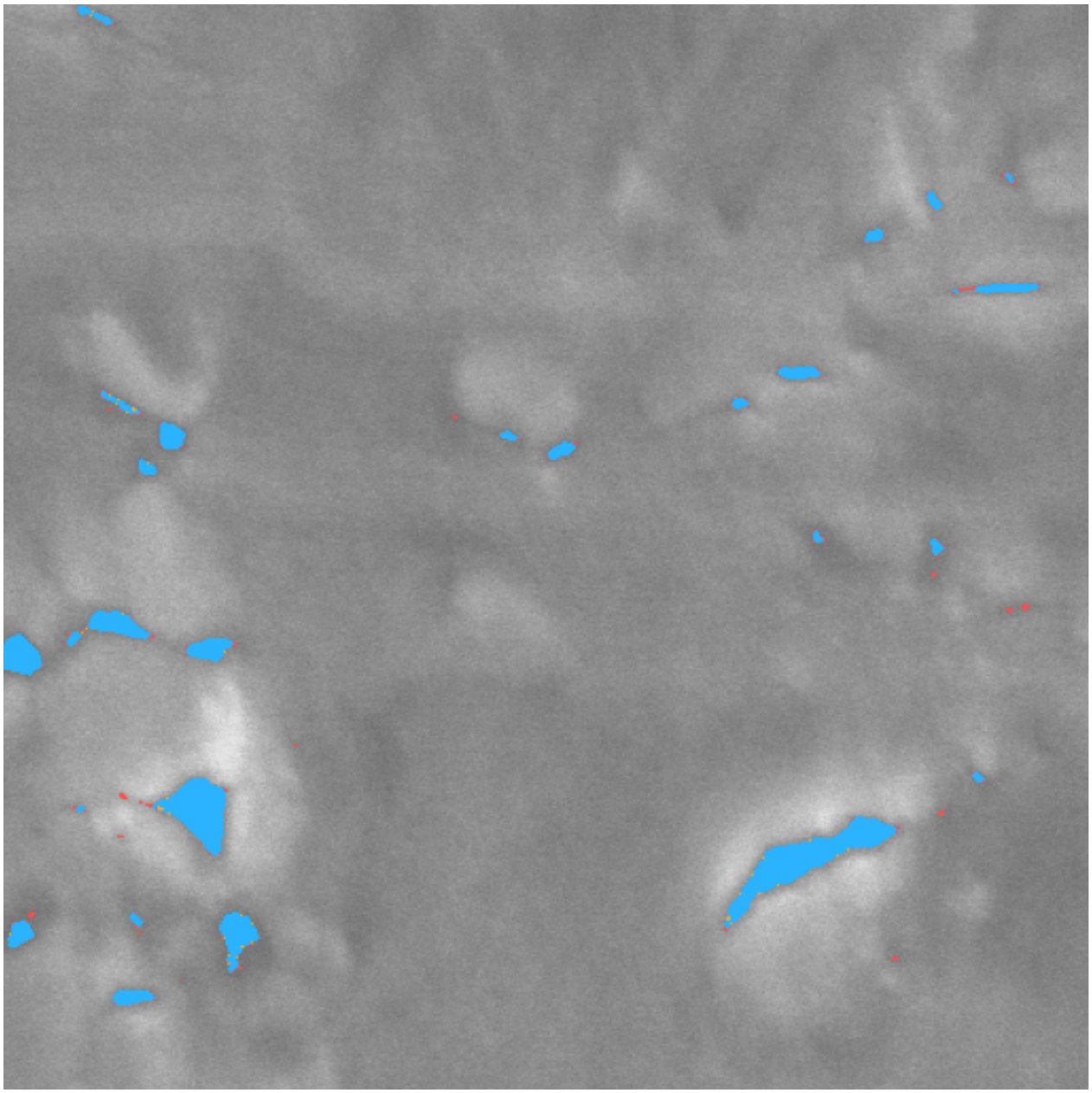


carries 3,889 px in this crop (1.48% of it) ·
667,935 px in the whole frame

7. clean_mask

What it does. Drops components of 13 px or fewer, opens with a disk of radius 1, closes with a disk of radius 3, fills holes.

What it costs here. removes 168 px, adds back 42 (the closing and the hole fill are extensive), net -126 px in this crop.



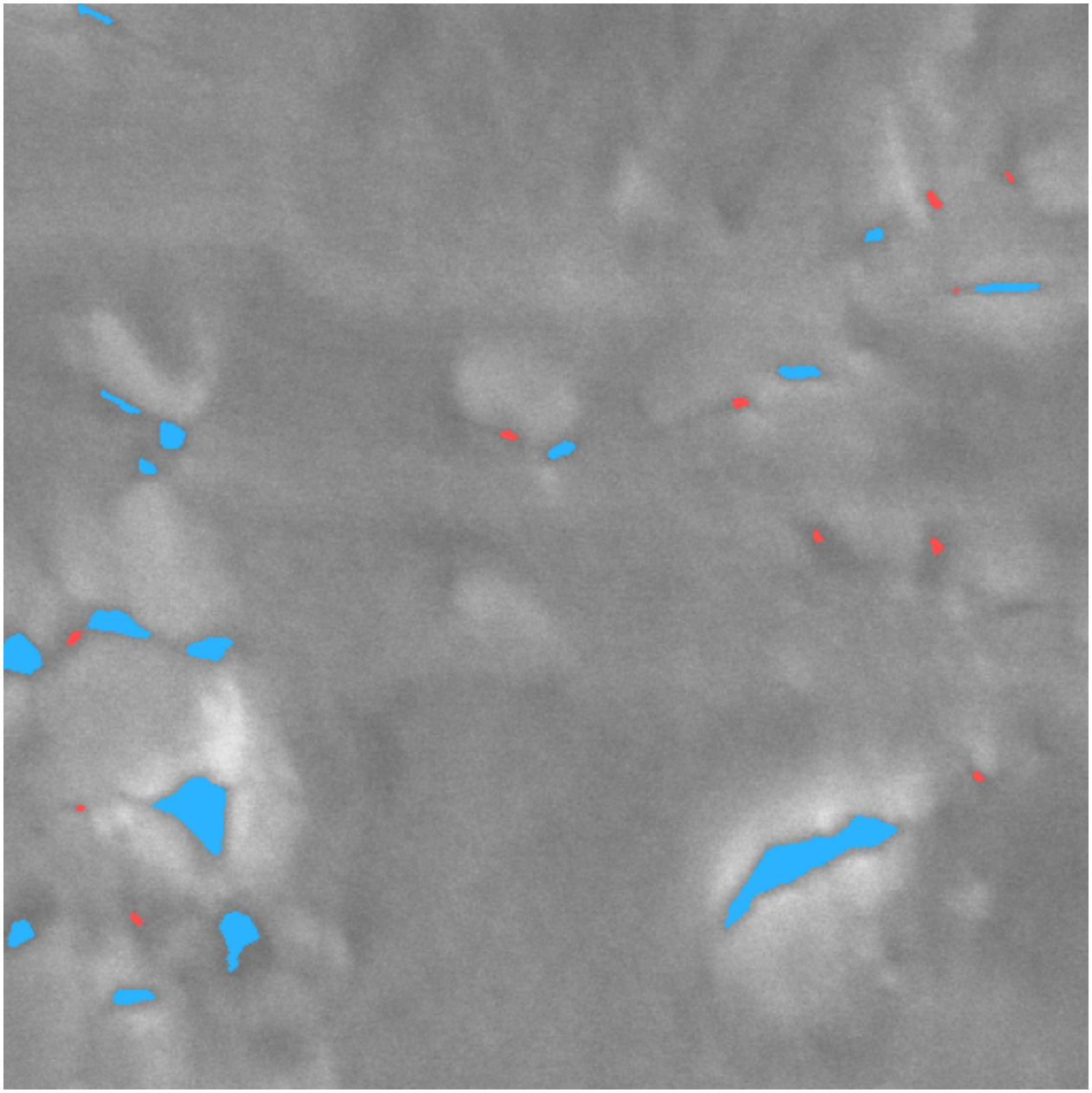
carries 3,763 px in this crop (1.44% of it) ·
666,796 px in the whole frame

8. The 40 px region floor

What it does. extract_candidates labels the clean mask with connectivity=2 and keeps only components of at least 40 px, renumbering the survivors 1..N. 465 candidate regions survive on this frame.

What it costs here. 249 px in this crop.

And what does not happen next. The region classifier is USE_REGION_CLASSIFIER = False, so every one of those regions is kept. Deleting it won on all four metrics with zero free parameters; six region-level precision rules tried since all scored well on 512 px crops and then deleted 54–75% of hand-painted crack on the 48 full frames.

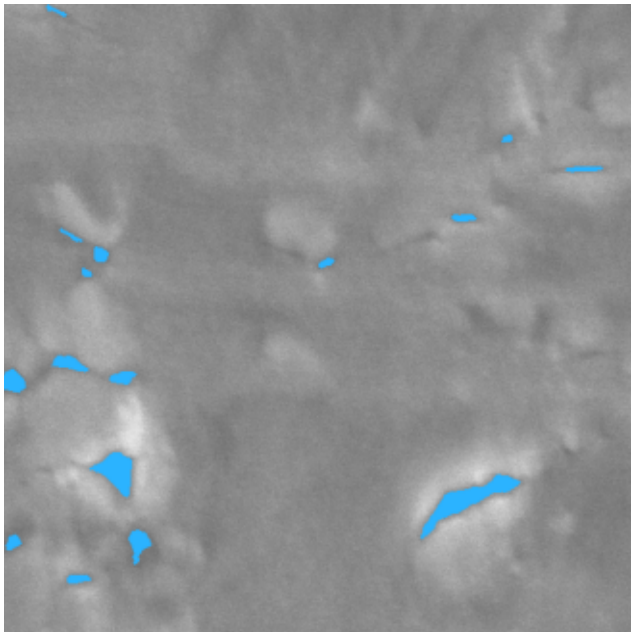


carries 3,514 px in this crop (1.34% of it) ·
662,941 px in the whole frame

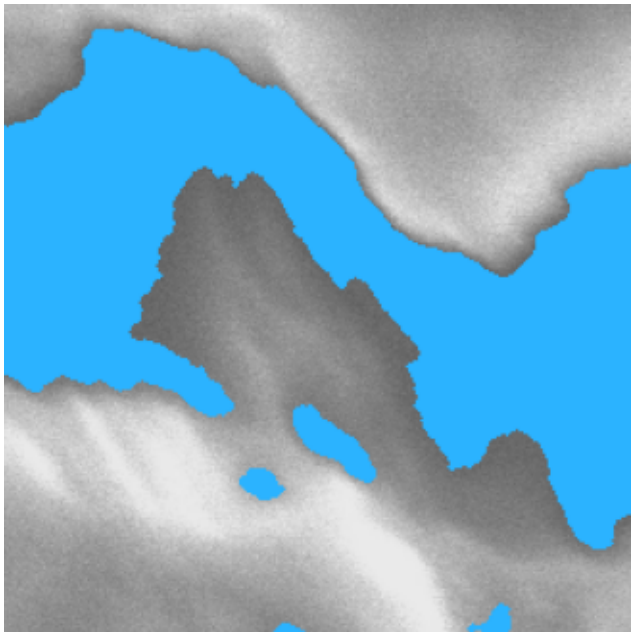
9. Pass 2

What it does. With a crack mask now in hand, proposes concavity, bridge-corridor and interior-fill candidates relative to it, scores them with the unified model, and folds the accepted ones in. 154 proposed on this frame, 129 accepted.

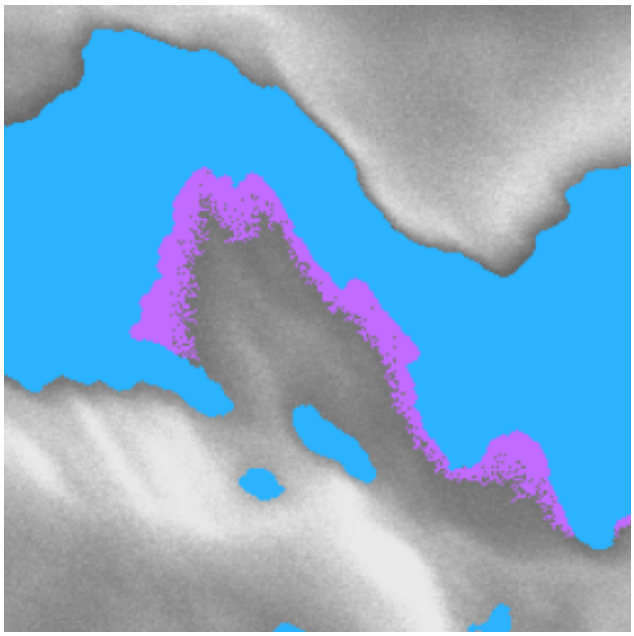
What it buys. +41,369 px frame-wide (5.87% of the final mask) and 0 px in this crop. The second pair of panels is a different place, not a different stage, and is shown because a panel of the main crop alone would read as the stage doing nothing anywhere.



carries 3,514 px in this crop (1.34% of it) · 704,310 px in the whole frame



A different 256 px window, at x=0 y=992: the mask before Pass 2.



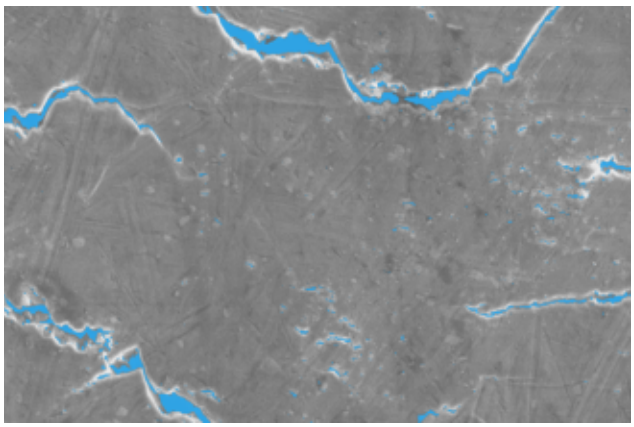
The same window after. Violet is Pass 2's addition — its largest accepted region is 3,273 px.

10. The emitted mask

What it does. Unions every kept region into one boolean mask and saves it. Area fraction 2.799% on this frame, 1.341% in this crop.



carries 3,514 px in this crop (1.34% of it) · 704,310 px in the whole frame



The same mask over the whole frame, for scale.

11. The measured output

What it does. Re-labels the emitted mask and measures every component: area, tip-to-tip geodesic, skeleton length, mean and max width, tortuosity, and whether it runs off the frame edge.

What it costs. 452 rows on this frame, 16 in this crop. Length is a bracket, not a value: 2.65% of regions on this frame touch the edge, but they are 29.57% of the crack area and 20.43% of the length. No scale bar was recovered for this frame, so every column here is pixels, not microns.



One colour per measured crack; grey is a component at or below the 25 px speck floor, which the measurement step drops. 16 components in this crop, 0 of them specks.

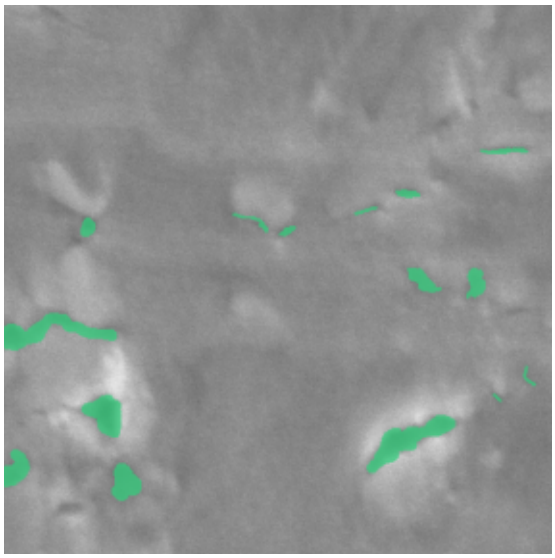
12. Against the hand paint, and where the recall goes

On this crop: precision 0.8557, recall 0.4346, IoU 0.4049. One crop is not a corpus figure.

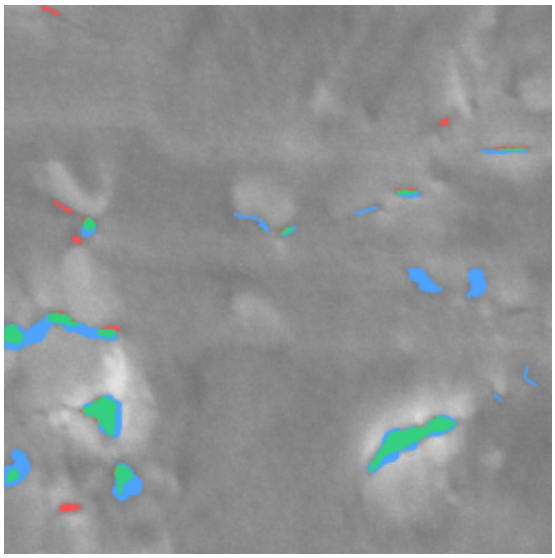
The two stages worth marking are the first row and everything else. On this crop the darkness cut owns 95.19% of all missed paint and the two morphology/area filters own 4.81% between them; over the 48 hand-painted full frames the same decomposition reads 96.02% and 3.98%. Tuning the morphology cannot reach the problem.

Reproduce every panel and every number on this page: make_flowchart.py beside this file, run under the detector environment. It asserts its own ladder against the pipeline's output and asserts the loss buckets are exhaustive before it writes anything.

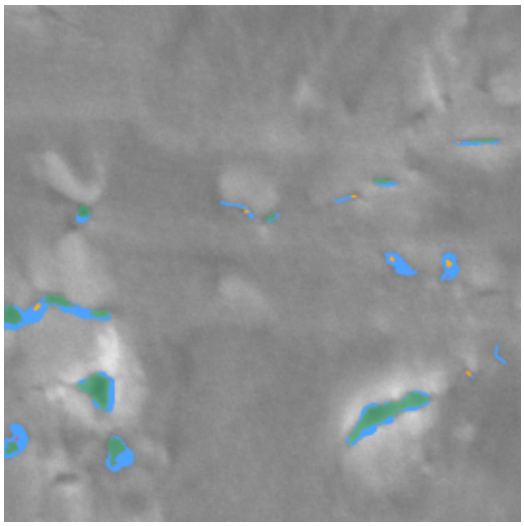
Note. How to read any number on this page. These are one crop of one frame, shown because the stages are comparable on it. The corpus figures, over the 26 hand-labelled tiles, move further with the aggregation than with any method change ever measured in this project, so a score is unreadable without its row and its tolerance:aggregationprecisionrecallF1IoUpooled over all pixels0.41820.62520.50120.3344median over the 26 tiles0.80560.76230.59220.4208median over 13 specimens0.78970.71590.54180.3715median over specimens, 2 px tolerance0.89880.80250.66330.4962A perfectly placed trace caps at IoU 0.4993 pooled and 0.5498 median over tiles on these labels, so 0.4208 is 84% of what the labels permit, not 42% of perfect. IoU and clDice are both width-gameable here — a no-information dilation to 1.5× area moves clDice +0.0478 on 13 of 13 specimens — so the metric to compare methods on is skeleton-F1 at a 2 px tolerance, where this detector scores 0.5507.



The hand paint for this crop: 6,919 px, 2.64% of it. Painted from blank; the detector had no hand in it.



green = found, red = marked but not painted, blue = painted but missed.



The missed paint, charged to the stage that dropped it: blue = never dark, red = clean_mask, amber = the 40 px floor.